IBM AML Dataset — Overview

Reference sheet for Week 2: Data Science Foundations & Domain Research

1. What the dataset is

The IBM Transactions for Anti-Money Laundering (AML) dataset is a fully synthetic dataset generated by IBM's multi-agent financial simulator. It models transactions between individuals, companies, and banks, and embeds labeled money-laundering activity alongside legitimate transactions. Because it is synthetic, it can be shared and used freely for research and training without any real customer data or privacy concerns.

Source: kaggle.com/datasets/ealtman2019/ibm-transactions-for-anti-money-laundering-aml

2. How the files are named

Each file follows the pattern: **[Ratio]-[Size]_[FileType]**

Ratio — HI vs LI

- **HI** = High Illicit ratio → more laundering transactions relative to legitimate ones
- **LI** = Low Illicit ratio → fewer laundering transactions relative to legitimate ones

Size — Small / Medium / Large

Refers to the overall volume of transactions and accounts covered (number of simulated days, accounts, and transactions). Small is the most practical for exploratory work on a laptop.

	Small	Medium	Large
Days spanned	10	16	97
Accounts (approx.)	515K	2,077K	2,116K
Transactions (approx.)	5M	32M	180M
Laundering transactions (approx.)	3.6K	35K	223K

Note: HI and LI variants at the same size have similar transaction/account counts but different laundering rates — HI-Small has roughly 1 laundering transaction per 981 total; LI-Small is roughly 1 per 1,942 (about twice as sparse).

3. File types inside each variant

- **[Name]_Trans.csv** — the main transaction table. One row per transaction: timestamp, sender account, receiver account, amount, currency, payment format, and an **Is Laundering** label (0/1). This is the primary file used for EDA.
- **[Name]_Patterns.txt** — lists which specific transactions belong to a known laundering pattern group (e.g. fan-in, fan-out, cycle, gather-scatter). Not all laundering transactions in the data follow one of these known patterns. Useful for the typology research task, not required for basic EDA.

4. Columns in Trans.csv

Each row in a *_Trans.csv file represents a single transaction between two accounts. The columns are the same across all HI/LI and Small/Medium/Large variants:

Column	Type	Meaning
Timestamp	object (string)	Date/time of the transaction; convert to datetime before analysis
From Bank	int64	ID of the bank the money is sent from
Account	object	Sender account identifier
To Bank	int64	ID of the bank the money is sent to
Account.1	object	Receiver account identifier
Amount Received	float64	Amount credited to the receiving account
Receiving Currency	object	Currency of the received amount
Amount Paid	float64	Amount debited from the sending account
Payment Currency	object	Currency of the paid amount
Payment Format	object	Payment method/rail (e.g. wire, ACH, credit card, cheque)
Is Laundering	int64	Label: 1 = laundering transaction, 0 = legitimate

Note: sender/receiver amount and currency are recorded separately (Amount Paid / Amount Received) because cross-currency transactions can differ slightly between the two sides.

5. Dataset chosen for this project

HI-Small_Trans.csv was selected as the working dataset for Week 2.

Brief reasoning:

- Small enough (~5M rows) to load and iterate on quickly within the 2-day timeline
- High illicit ratio (HI) gives more laundering examples to analyze despite the smaller size
- Already contains real fan-out, fan-in, and cycle patterns needed for the typology research task
- If time allows, the same notebook code will be re-run on HI-Medium to validate that findings hold at larger scale